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EXP 2: PERFORM WEBSITE, EMAIL, WHOIS, DNS, AND NETWORK FOOTPRINTING ON THE TARGET NETWORK

AIM:

To gather detailed technical information about the target's website, email servers, domain registration (Whois), DNS records, and network range using footprinting tools.

THOERY:

This experiment moves one step further than Experiment 1 — here we look at the technical infrastructure of the target instead of just public/social information.

- Website Footprinting: Studying the structure, technology, and hidden files of a website.
- Email Footprinting: Finding out mail server details and tracing email headers.
- Whois Footprinting: Whois is a public database that stores information about who registered a domain name — owner name, registrar, registration date, and name servers.
- DNS Footprinting: DNS (Domain Name System) translates domain names into IP addresses. DNS records (A, MX, NS, TXT, CNAME) reveal a lot about a target's infrastructure.
- Network Footprinting: Finding the IP address range and network path (route) used by the target organization.

TOOLS REQUIRED:

1. whois command (Kali Linux)
2. nslookup and dig commands
3. traceroute command
4. Netcraft (https://www.netcraft.com) — website technology lookup
5. Wappalyzer / WhatWeb — technology fingerprinting
6. Web browser Developer Tools (View Page Source)
7. Target domain: rajalakshmi.org.

PROCEDURE:

Step 1: Whois Footprinting

In the Kali terminal, run:

```
whois targetcompany.com
```

Look for the following in the output:

- Registrar name
- Registrant name/organization (if not privacy-protected)
- Creation date and expiry date
- Name servers (NS records)

Step 2: DNS Footprinting

Use nslookup to find the IP address of the domain:

```
nslookup targetcompany.com
```

Find the mail server (MX records):

```
nslookup -type=mx targetcompany.com
```

Use dig for a more detailed view, including all record types:

```
dig targetcompany.com ANY
dig targetcompany.com NS
dig targetcompany.com TXT
```

If a zone transfer is misconfigured (rare, but useful to check in lab environments), try:

```
dig axfr @nameserver targetcompany.com
```

Step 3: Website Footprinting

Open the target website and view the page source (Ctrl+U in browser) to check:

- Comments left by developers in the HTML
- JavaScript library names and versions (old versions may have known vulnerabilities)
- Hidden form fields

Also run WhatWeb from the terminal to fingerprint web technologies:

```
whatweb targetcompany.com
```

This shows the web server type (Apache/Nginx/IIS), CMS (WordPress/Joomla), and programming language used.

Step 4: Email Footprinting

Send yourself a test email from a contact form on the target site (only in a lab setup where

this is permitted), and open the received email's full headers. Email headers reveal:

- The sending mail server's IP address
- The software used to send the mail (e.g., Exchange, Postfix)

Step 5: Network Footprinting

Find the network path to the target using traceroute:

```
traceroute targetcompany.com
```

This shows every router (hop) the traffic passes through before reaching the target, which helps in understanding the target's network structure and possibly identifying firewalls or load balancers.

OBSERVATION TABLE:

INFORMATION	RESULT
Target Domain	rajalakshmi.org
WHOIS (Using ICANN Lookup)	Registrar: Network Solutions, LLC; Domain status: clientTransferProhibited; DNSSEC: Unsigned; Creation date: 2003-01-08 05:42:32 UTC Expiry date: 2027-01-08 05:00:00 UTC
DNS Records (Using Google Admin Toolbox Dig)	TTL:15 mins DATA: 192.64.115.115 MX record :No public MX record
Technologies (Using BuiltWith)	Next.js; Google Analytics; Microsoft Clarity; Google Tag Manager
Subdomains (Using crt.sh)	www.rajalakshmi.org, architecture.rajalakshmi.org, admissions.rajalakshmi.org

RESULT:

Footprinting of rajalakshmi.org was performed using public lookup tools. Domain details, DNS information, website technologies, and subdomains were collected and documented.